GROUP WORK PROJECT #FE GWP 3 M7 MScFE 610: FINANCIAL DATA

GROUP NUMBER: 7681

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

Modeling non-stationarity and finding an equilibrium

Definition – Vector Error Correction Model

Equation (MscFE 610 Financial Econometrics):

$$.\Delta X_t = C + \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \varepsilon_t$$

Where:

C is the deterministic vector or matrix depending on if there are constants and/or linear trend.
Π is a 3X3 matrix for coefficients of lagged X_{t-1} vector.
Γ_i is a 3X3 matrix for coefficients of lagged differences of X.
ϵ_t is a white noise 3X1 vector with means and stable covariance.

Vector error correction (VEC) models are related to vector autoregressive models (VAR), the major difference between a VEC model and a VAR model is the error correction term: ΠX_{t-1}

Error correction term: In an ECM, the error correction term represents the extent to which the previous period's disequilibrium influences the current period's adjustments. This allows the model to capture both short-term fluctuations and the speed at which the variables return to equilibrium. (Geeksforgeeks, 2024)

It is widely used to analyze relationships among economic variables, such as exchange rates, interest rates, GDP components, and asset prices.

Cointegration Rank: The concept of cointegration is basically to see the long-term balance among the observed variables. Sometimes a data that is individually is not stationary, but when linearly connected the data becomes stationary. It is then called that the data is cointegrated. In the Johansen test, the determination of cointegration is seen from the value of trace statistic and max eigen statistic after preceded by finding the length of the lag to be known. Trace statistic and max eigen statistic values that exceed their critical values indicate that there is cointegration in the model used. (Agus Suharsono; Auliya Aziza, 2017)

Number of lagged differences (K_{AR-1}): The number of lags to be used in the model can be determined based on the criteria of Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Final Prediction Error (FPE) or Hannan-Quinn Information Criterion (HQIC).

Description

The VEC model structure is used for multivariate cointegration analysis and can be used to analyze both long-term and short-term dynamics of cointegrated nonstationary $I(1)$ time series variables.

The dynamics of the model is used to implement a long-run change by changing the current values. The counterfactual experiment can be conducted precisely when the cointegrating relation is identified. (Johansen, 2005)

Demonstration

To retrieve historical price data, yahoo finance was used. The dataset consists of daily price series of BTC-USD, ETH-USD, SOL-USD. The crypto market is open 24/7 hence the time series does not contain price gaps even though we must check whether the data contains nan or missing values. The fetched data covers the daily adjusted close prices from 2015 January 1st to 2023 December 31st.

During the selection of dataset we kept non-stationary and cointegration test results in focus, so that we can work with the most suitable dataset.

The first tests we apply to our time series are the Augmented Dickey Fuller (ADF) Test and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test to check if the data is stationary. From the returned p-values we can conclude that the raw price data is non-stationary. Next, we apply the tests to the first differences which show non-stationary as well. To continue with the VEC model we turn to log values and run the tests again. The log level values show non-stationary, whilst the first difference log values show stationary p-values. Now that we meet with the model's assumption for stationary, we can go on with the cointegration test.

For the cointegration test we turn to the Johansen-test, measuring the Trace and Eigen returns we reject the null hypothesis and assume that cointegration is present. To define the cointegration rank for our model, we check the critical and test statistical values. In this test at a 5% critical level, we decide to use $r = 1$ as cointegration rank since $r = 2$ has weaker evidence. With this the model's assumption for cointegration is present we can proceed with applying the VECM.

To apply the model, we need to select the optimal lag which we decided to do with the AIC method, for a more standardized approach.

Once the lags have been selected, we can fit the model.

Diagram

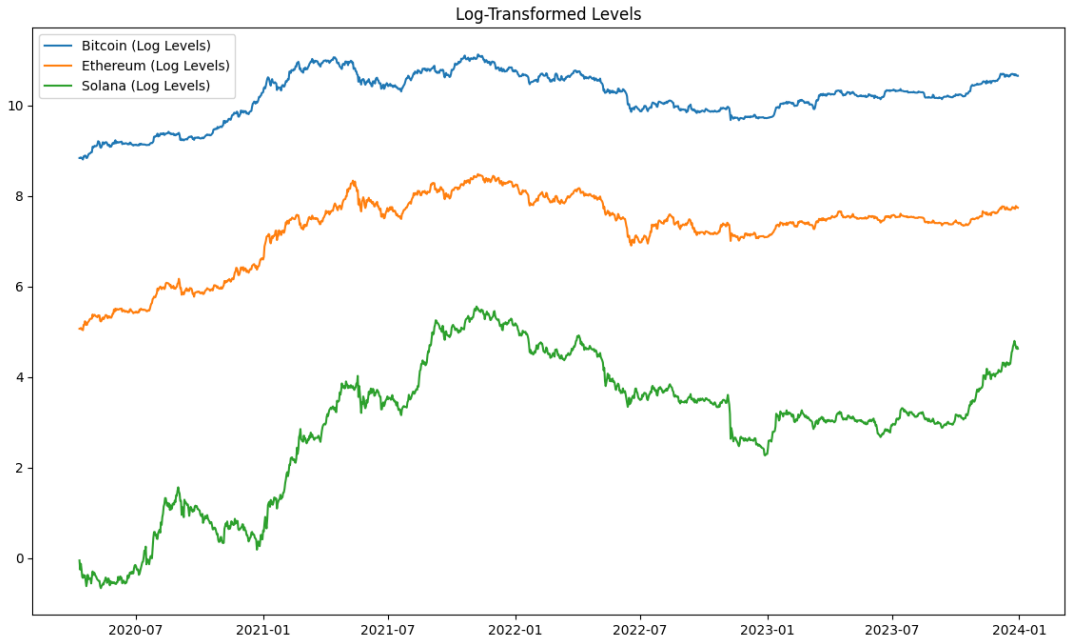

Figure 1

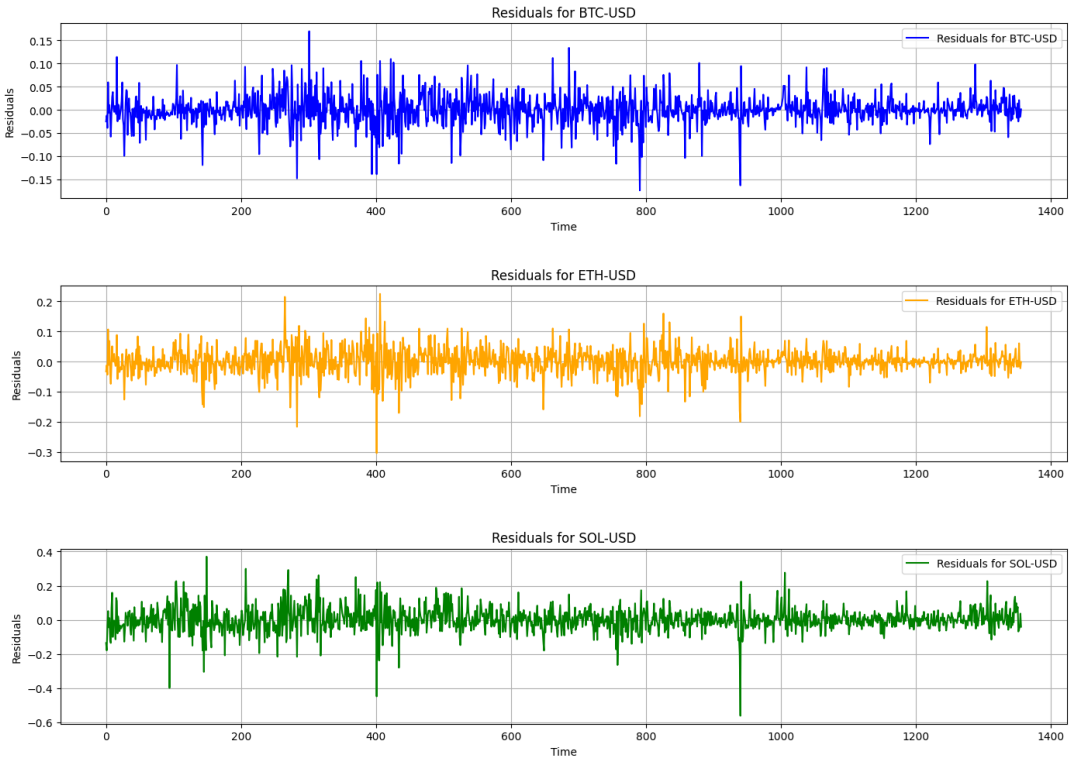

Figure 2

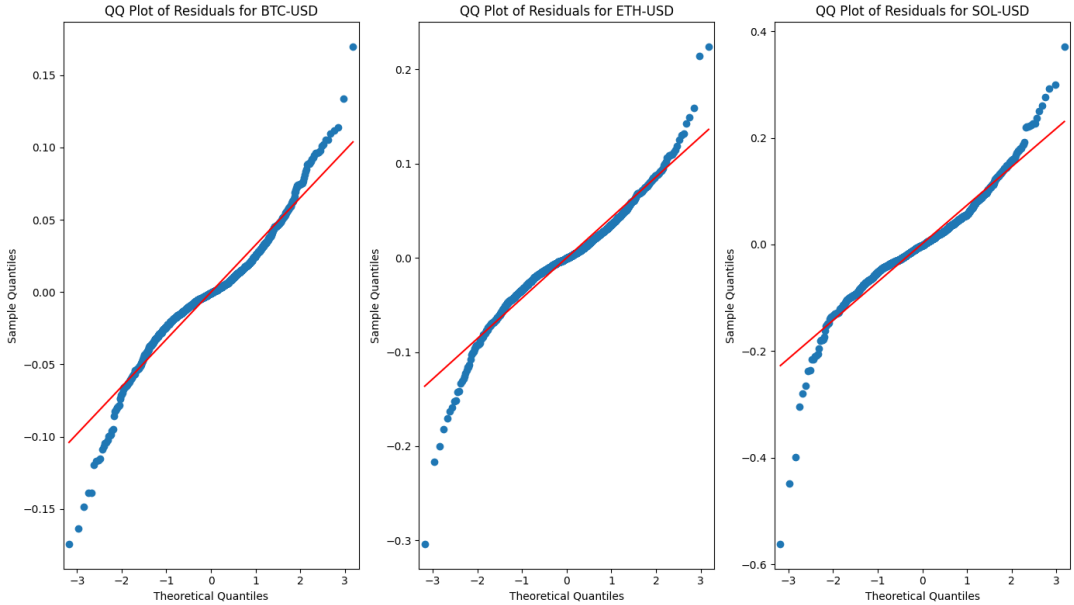

Figure 3

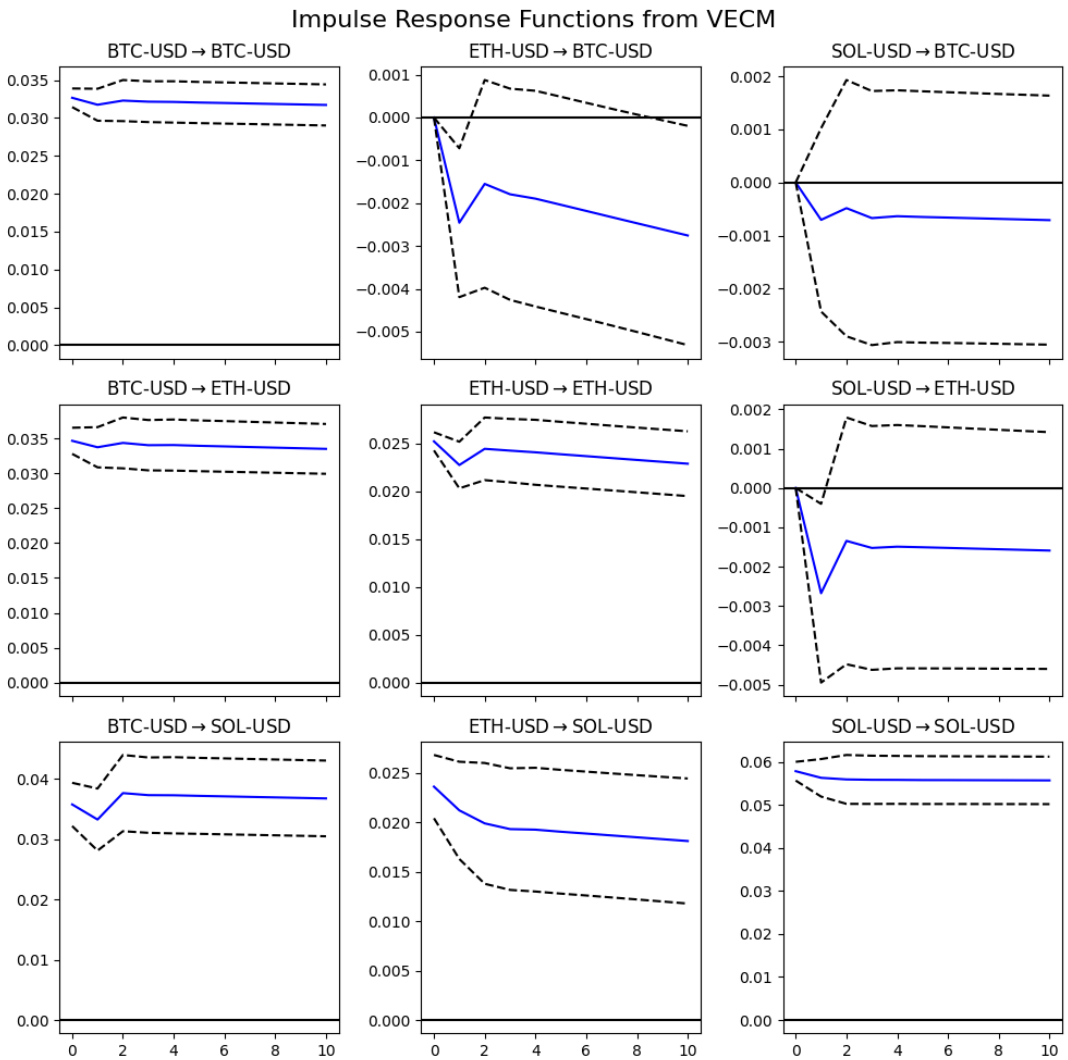

Figure 4

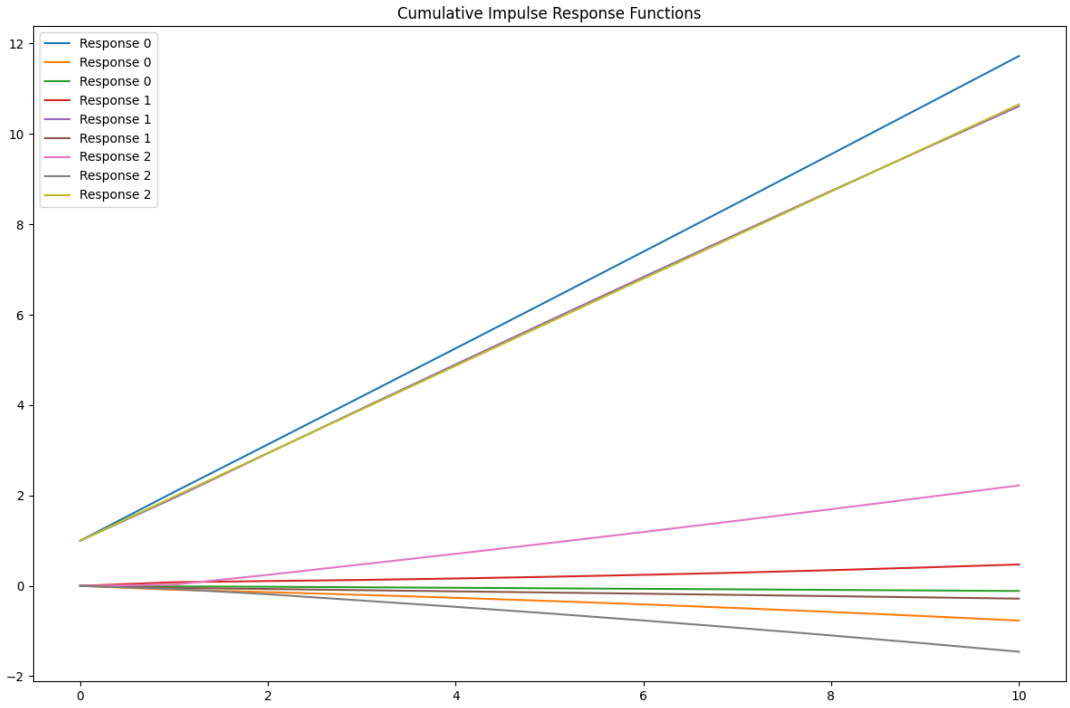

Figure 5

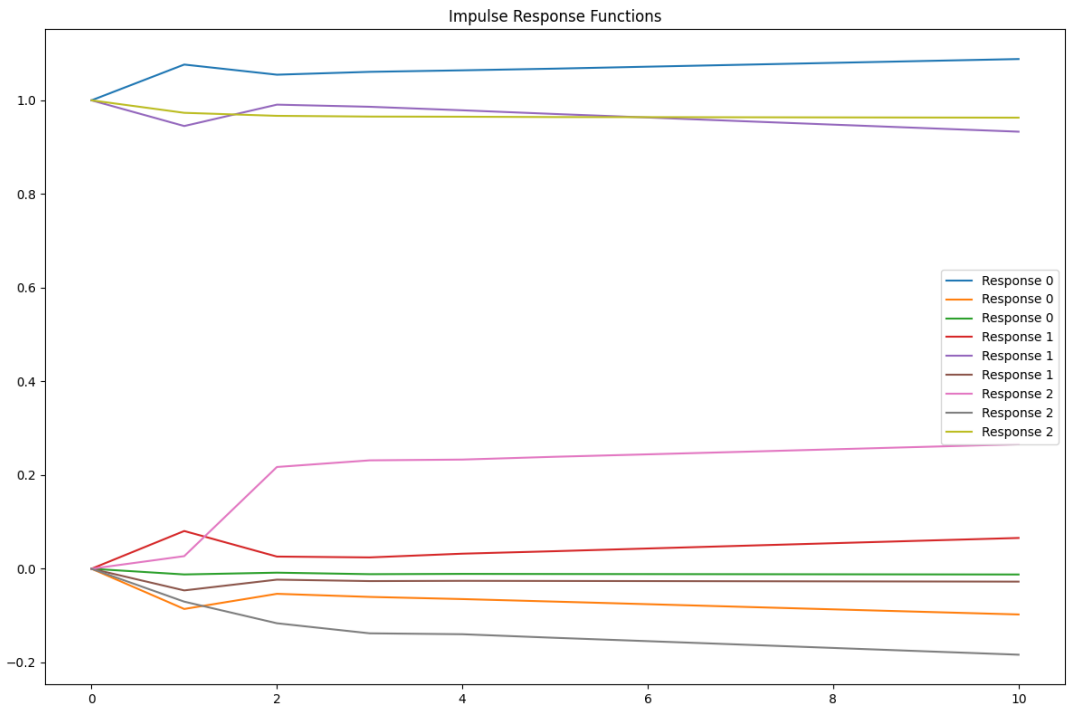

Figure 6

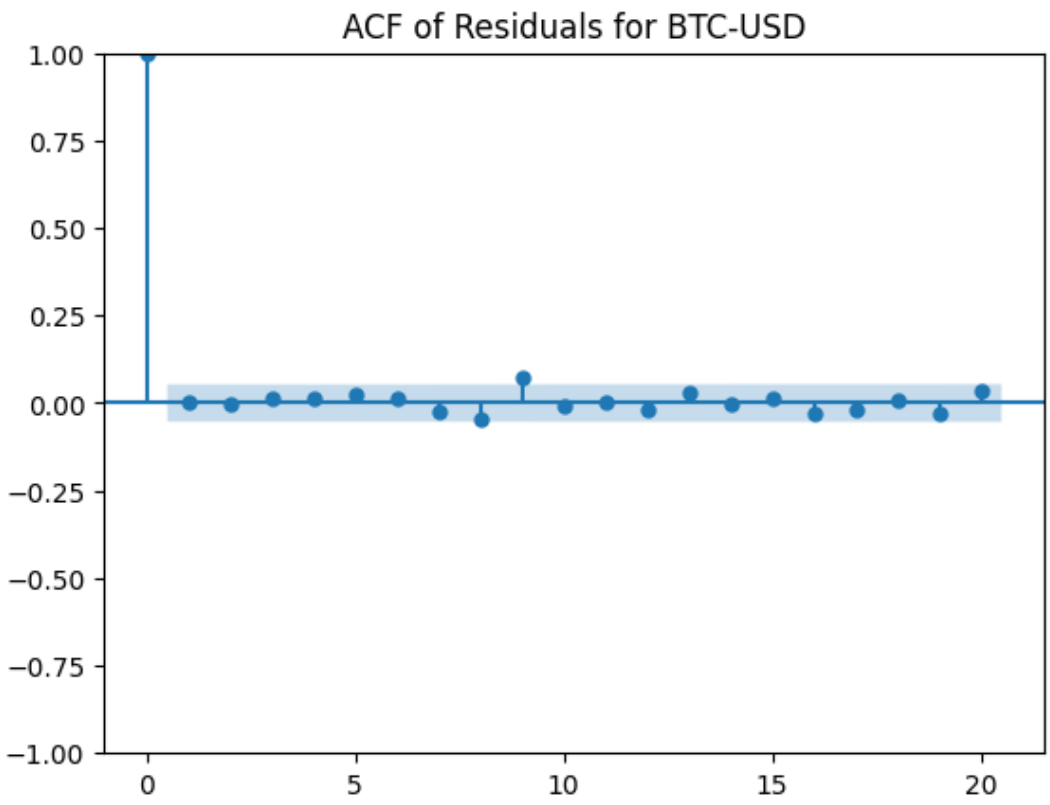

Figure 7

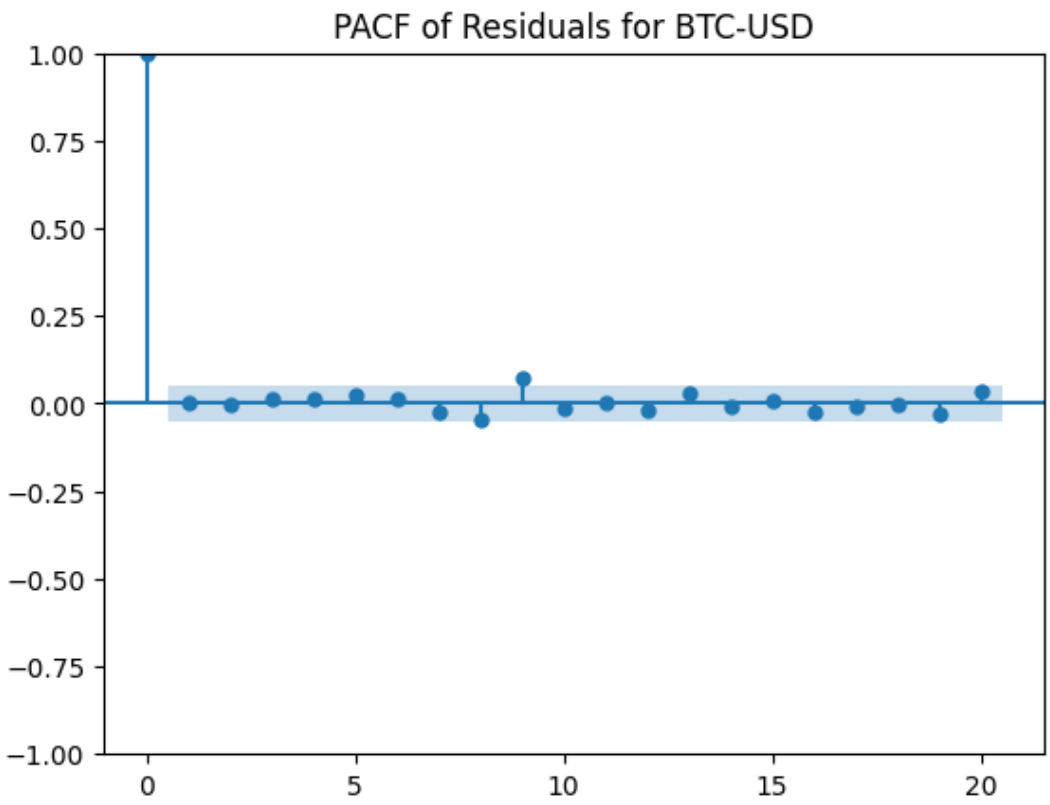

Figure 8

Diagnosis

The equation of VECM the cointegrating vector coefficients β is:

$\beta 1 = 1.000$ (BTC)

$\beta 2 = -1.324$ (ETH)

$\beta 3 = -0.077$ (SOL)

The coefficients describe the long-term equilibrium relationship between the cryptocurrencies. the equation is:

$$BTC - 1.324 \times ETH - 0.077 \times SOL = 0$$

Looking at the equation we can tell that in long term relationship, the price movements in Bitcoin are neutralized by changes in Ethereum and Solana. When this larger coefficient for Ethereum (-1.324) is compared to Solana (-0.077) it proves that Ethereum helps in the equilibrium relationship.

The loading coefficients (α) in the model are BTC is 0.0042 and for ETH:is 0.0057

These coefficients show that how fast these cryptocurrencies adjust when the system moves away from its long-term position. The positive values tell us that all three cryptocurrencies restore the equilibrium when disrupted.

For example, take a scenario where the deviation from equilibrium increases by 1%. then in the next period:

Bitcoin will adjust by about 0.42%

Ethereum will adjust by about 0.57%

Solana will adjust by about 0.54%

These are all statistically significant as provided in the code. Ethereum shows the fastest adjustment rate (0.0057), then Solana (0.0054), where Bitcoin shows the slowest adjustment (0.0042). This could make senses as Bitcoin's role as the market leader - it tends to be more resistant to adjusting to movements in other cryptocurrencies.

The small size of the coefficients (which is less than 1%) is good for the model as the adjustment process is slower than fast and could be an opening for creating arbitrage opportunities. The system maintains stability through these small but consistent corrections.

These cryptocurrencies have got good adjustment coefficients and having a robust equilibrium relationship. Each cryptocurrency responds to deviations from the long-term relationship, though they do so at a smaller pace with different speeds. These markets do diverge in the short term, but they tend to move back toward their original position over time.

The upward-trending cumulative effects (visible on Figure 5, particularly the strong blue line reaching 12 units) demonstrate that cryptocurrency shocks create permanent price level changes rather than temporary fluctuations. This provides the investors about how we should treat major market events as regime changes require portfolio rebalancing, rather than opportunities for mean-reversion trading strategies. The varying slopes between

cryptocurrencies also indicate different speeds of market adjustment, which creates potential arbitrage opportunities.

Short-term market dynamics reveal that cryptocurrency price shocks primarily affect their own markets, in which we see that immediate trading responses should focus on the impacted currency. Moderate cross-market effects give us the opportunities for short-term arbitrage between related cryptocurrencies.

Damage

Misleading Inferences: Non-normal residuals and autocorrelation present. The Shapiro-Wilk and Jarque-Bera tests indicate that the residuals are not normally distributed. This violates assumptions required for many statistical tests, leading to invalid confidence intervals and hypothesis tests.

Inefficient Estimation: Cause is Autocorrelation in residuals. As Ljung-Box test shows significant autocorrelation in the residuals, which hampers the model to capture the data dynamics .and leads to inefficient parameter estimates like the Parameter estimates may have larger variances, reducing the model's precision in predictions.

ARCH Effects: The presence of ARCH effects indicates time-varying volatility, this is a common feature in financial time series. This can lead to unreliable forecast and risk assessment.

Misspecification in Lag Selection and Cointegration Rank:

Incorrect lag length or cointegration rank can lead to flawed inferences, affecting the model's overall validity.

Directions

Outlier Detection: Remove or adjust extreme values in SOL-USD.

Variance Stabilization: apply transformations (e.g., log or Box-Cox) to stabilize variance across series.

Shorten Time Horizon: Using a time window where cointegration and relationships are more robust.

Experiment with higher lag in orders to better capture dynamics (beyond the optimal lag from AIC). or use BIC.

Use dummy variables or interaction terms to see for structural breaks or market events.

As ARCH test shows heteroskedasticity we need to explore modeling GARCH-VECM model to consider the volatility clustering basically to jointly estimate conditional volatility and mean dynamics.

Scaling and Normalization: Use standardization techniques for example z-scores

Tracking Cointegration: Instead of assuming fixed cointegration coefficients, we can use the Kalman Filter to estimate time-varying coefficients (Wang et al, 2012)

Deployment

Calculating optimal position sizes based on cointegration coefficients we can maintain balanced exposure. For example, hold 1 BTC, the equilibrium relationship tells us to hold 0.75 ETH (1/1.324) and about 13 SOL (1/0.077) to keep normal exposure. But these positions should be adjusted based on risk tolerance and market conditions. This should be taken with a volatility management layer where volatility changes can be taken up by dynamic risk limits. During high volatility periods, position sizes should be automatically reduced. The direction would be to create volatility bands.

Forecasting: We can utilize the error correction term to predict short-term price adjustments toward equilibrium. The deviations in BTC-USD from its equilibrium with ETH-USD could signal mean reversion.

Trading Strategies: mean-reversion strategies that capitalize on deviations from the cointegration relationship. For instance:

- When BTC-USD price is above equilibrium, short BTC and long ETH.
- Monitor residuals and standardize thresholds for actionable trading signals.

Portfolio Optimization: Using cointegration results to hedge portfolios by combining BTC, ETH, and SOL in proportions that minimize risk while maintaining long-term equilibrium.

Computational Resources: Must ensure and assess the computational requirements for real-time processing, especially for high-frequency trading strategies in the requirements low latency requirements.

Data Pipeline: Automating data ingestion and preprocessing using Python tools like pandas or package API for model deployment on cloud ready to be used can be thought where:

- Log transformation and differencing.
- Outlier detection and adjustment.
- Real-time updates for deploying the model.

Regulatory compliance: Must ensure that the model's deployment complies with all relevant laws and regulations in the target jurisdictions.

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